

Experiment No: 07

Date: 17/09/2026

Title: Bank Interest Rate System using Inheritance and Method Overriding

Objective / Aim

To design and implement a Bank Interest Rate System in Java using inheritance and method overriding, demonstrating the concepts of classes, objects, inheritance, method overriding, and runtime polymorphism.

System Requirements

Hardware

- Standard PC/Laptop
- Intel Core i3/i5 processor or equivalent
- Minimum 4 GB RAM
- Minimum 500 MB free storage

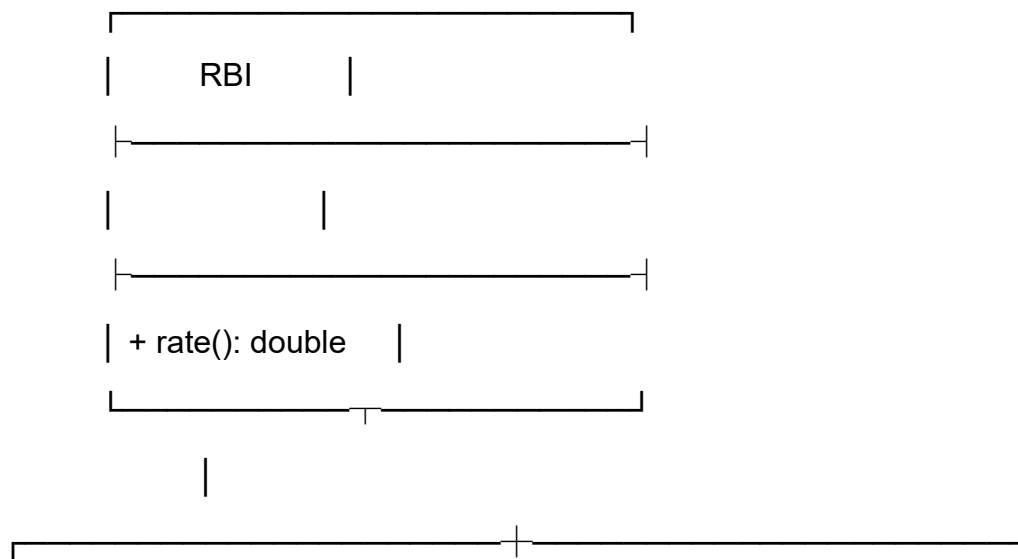
Software

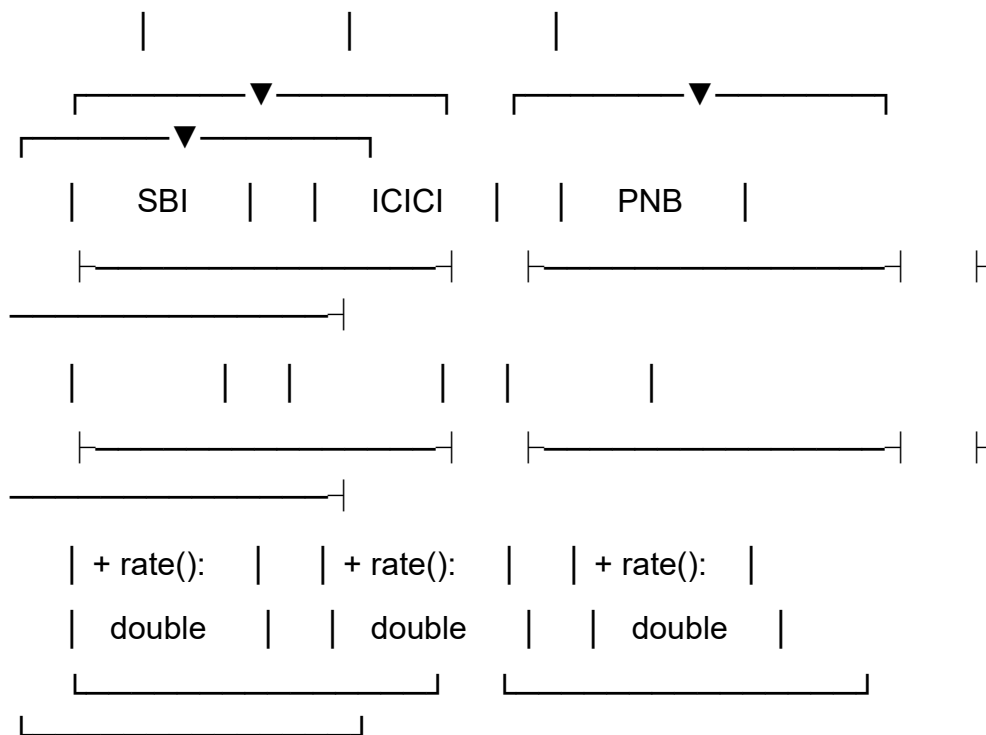
- Operating System: Windows / Linux / macOS
- Java Development Kit (JDK 17 or JDK 21)
- Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
- Command Prompt / Terminal

Algorithm / Logic Description

1. Create a parent class named RBI.
2. Define a "rate()" method in the RBI class to return the minimum interest rate.
3. Create an SBI class that inherits from the RBI class.
4. Override the "rate()" method in the SBI class to return the SBI interest rate.
5. Create an ICICI class that inherits from the RBI class.
6. Override the "rate()" method in the ICICI class to return the ICICI interest rate.
7. Create a PNB class that inherits from the RBI class.
8. Override the "rate()" method in the PNB class to return the PNB interest rate.
9. Create a "Scanner" object to accept the bank name from the user.
10. Create an RBI parent-class reference.
11. Create the appropriate bank object based on the entered bank name.
12. Call the overridden "rate()" method using the RBI reference.
13. Display the corresponding bank's rate of interest.

UML Class Diagram





Relationship: "SBI", "ICICI", and "PNB" inherit from the "RBI" class. The "rate()" method is overridden in each child class.

Source Code

```

package javacore;

import java.util.*;

// Parent class RBI
class RBI {

    // Method to return minimum interest rate
    double rate() {
        return 4;
    }
  }

```

```
}  
}
```

```
// SBI inherits RBI
```

```
class SBI extends RBI {
```

```
    // Override rate method for SBI
```

```
    double rate() {
```

```
        return 7;
```

```
    }
```

```
}
```

```
// ICICI inherits RBI
```

```
class ICICI extends RBI {
```

```
    // Override rate method for ICICI
```

```
    double rate() {
```

```
        return 6;
```

```
    }
```

```
}
```

```
// PNB inherits RBI
```

```
class PNB extends RBI {
```

```
    // Override rate method for PNB
```

```
    double rate() {
```

```
        return 5;
```

```
    }
```

```
}
```

```
// Main class

public class Bank {

    public static void main(String args[]) {

        // Create Scanner object
        Scanner sc = new Scanner(System.in);

        // Ask the user to enter bank name
        System.out.printf(
            "Enter the Bank name to find the rate of Interest : "
        );

        String s = sc.next();

        // Create parent class reference
        RBI b;

        // Create object based on bank name
        if (s.equalsIgnoreCase("RBI"))
            b = new RBI();

        else if (s.equalsIgnoreCase("SBI"))
            b = new SBI();

        else if (s.equalsIgnoreCase("ICICI"))
            b = new ICICI();
```

```
else
    b = new PNB();

    // Call overridden method using parent reference
    System.out.printf(
        "RBI rate of interest is : %.0f%%",
        b.rate()
    );
}
```

Compilation & Execution Commands

Save the source code as:

Bank.java

If you are using the "package javacore;" statement, save the file inside the appropriate "javacore" package/folder.

Compile the program:

```
javac Bank.java
```

Run the program:

```
java Bank
```

Output Screen

Example 1 – SBI

Enter the Bank name to find the rate of Interest : SBI

RBI rate of interest is : 7%

Example 2 – ICICI

Enter the Bank name to find the rate of Interest : ICICI

RBI rate of interest is : 6%

Example 3 – PNB

Enter the Bank name to find the rate of Interest : PNB

RBI rate of interest is : 5%

Example 4 – RBI

Enter the Bank name to find the rate of Interest : RBI

RBI rate of interest is : 4%

Result

The Bank Interest Rate System was successfully implemented in Java using inheritance and method overriding. The program demonstrates runtime polymorphism by using an RBI parent-class reference to call the overridden "rate()" method of different bank classes such as SBI, ICICI, and PNB.